

Homework 1: Regular Languages

Due: 1/15/2025

You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments. Three bonus points if you use L^AT_EX, and our template. You may collaborate with other students, but any written work should be your own.

1. One perspective of this class is that we are developing a theory of representation, or definability. For finite sets, this is easy, but it becomes more challenging for infinite languages. This class is really about the ability of different finite objects to understand and describe infinite sets.

- (a) Prove every finite language is regular.

Answer: We define L to be a finite language with an alphabet Σ . We wish to show L is regular. Since L is defined to be finite, we have the following:

$$L = \{w_0, w_1, \dots, w_k\}$$

where each w_i is a string over Σ .

Since each w_i is a string over Σ , we are able to separate L into distinct elements $\{w_i\}$.

$$L = \{w_1\} \cup \{w_2\} \cup \dots \cup \{w_k\}$$

We wish to now prove each set $\{w_i\}$ is regular by constructing a DFA or NFA (synonymous) for each w .

We begin by creating a starting state q_0 and for the string $w = a_1a_2\dots a_n$, where a_i are the characters of the string, we add a sequence of states such that from q_0 there is a transition on the symbol a_1 to q_1 . Similarly, from q_1 there is a transition on the symbol a_2 to q_2 . We repeat this process until the final state of q_n , where q_n is the only accepting state. Since the only string accepted by w is itself, we find $\{w\}$ is a finite automaton and therefore is a regular language.

As defined in lecture, we find regular languages are closed under the union operation which proves

$$L = \{w_0, w_1, \dots, w_k\} = \{w_1\} \cup \{w_2\} \cup \dots \cup \{w_k\}$$

is regular.

Since every finite language can be broken down as per the previously defined process above where every $\{w_i\}$ is regular, we have proven that regular languages are closed under union and therefore finite languages are regular.

- (b) Suppose we relax our definition of DFA too much. We define a DIA¹ to be exactly like a DFA except $|Q| = |\mathbb{N}|$, and δ, F are adjusted appropriately. Prove that *every* language $L \subseteq \Sigma^*$ is decided/accepted by a DIA.

Answer: We define the set of states to be all possible strings in Σ^* . For brevity, we will let $\Sigma = \{a, b, c\}$. We loosely define Q as the following:

$$Q = \Sigma^* = \{\varepsilon, a, b, c, aa, ab, ac, \dots\}$$

We begin by defining $q_0 = \varepsilon$. For each state $w \in Q$ and each input $a \in \Sigma$, we define a transition function to be $\delta(w, a) = wa$. In Layman's terms, if we are in state w and we see an input symbol a , we move to state wa . We repeat this process for all input symbols.

We wish for the automaton to accept only the language L , so we wish to define the accepting states as the following:

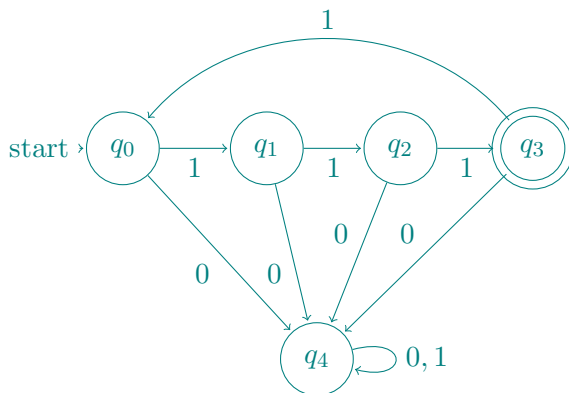
$$F = L \subseteq \Sigma^*$$

In Layman's terms (since symbols are confusing), we include F in all the states that are in L . Any state (or string) that is not in L is non-accepting.

Therefore, we have proven that for any language L over Σ , there is a DIA that recognizes L .

2. Give a DFA for the following languages. Your solution doesn't have to be minimal, but it may assist the TA in grading. Here $int(w, 2)$ converts the binary string w to a number. Assume that $\Sigma = \{0, 1\}$.

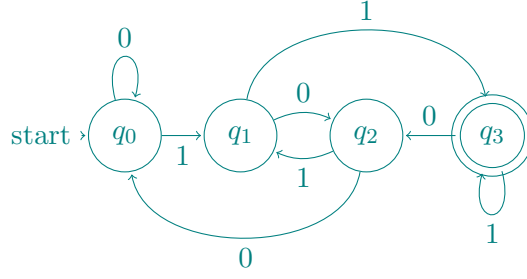
- (a) $L_1 = \{1^n \mid n \equiv 3 \pmod{4}\}$



Answer:

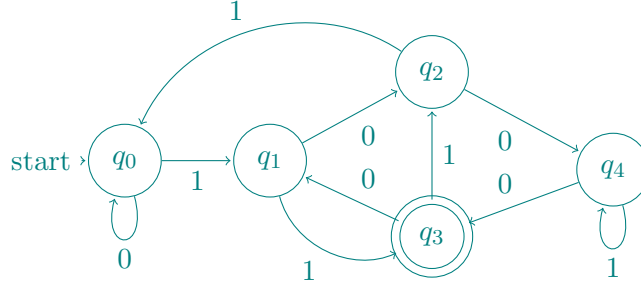
¹Deterministic Infinite Automata

(b) $L_2 = \{w \in \Sigma^* \mid \text{int}(w, 2) \equiv 3 \pmod{4}\}$



Answer:

(c) $L_3 = \{w \in \Sigma^* \mid \text{int}(w, 2) \equiv 3 \pmod{5}\}$



Answer:

3. Let L be a language from the alphabet $\Sigma = \{a, b\}$. Let L' be a language over the alphabet $\Sigma' = \{a, b, c\}$ where

$$L' = \{xcy \in \Sigma'^* \mid xy \in L\}$$

We take the strings in L , and add a single new symbol c anywhere in the string, even possibly at the beginning or end. Prove that if L is regular, then so is L' .

Answer: We define $M = (Q, \Sigma, \delta, q_0, F)$ to be an NFA for L . We wish to construct an NFA for M' for L' as follows:

- Set of States:** We wish to utilize $Q \times \{0, 1\}$ to track the state of M and if it has encountered c . A pair $(q, 0)$ represents being in state q and not yet having encountered c . Similarly, a pair $(q, 1)$ represents being in state q and having encountered c .
- Input Alphabet:** Trivial. We observe the input alphabet to be defined in the question.
- Transition Function:** For each pair (q, i) and input symbol σ , we have the following cases:
 - If $\sigma \in \{a, b\}$: $\delta'((q, i), \sigma) = \{(q', i) \mid q' \in \delta(q, \sigma)\}$
 - If $\sigma = c$ and $i = 0$: $\delta'((q, 0), c) = \{(q, 1)\}$
 - If $\sigma = c$ and $i = 1$: $\delta'((q, 1), c) = \emptyset$
- Initial State:** We trivially define the initial state to be $(q_0, 0)$ which indicates we have not yet encountered c .

- (e) **Accepting States:** We wish to accept strings that contain exactly one string and that, upon removing the c , lead M into an accepting state. Therefore, we define the accepting state to be $\{(q, 1) \mid q \in F\}$.

The NFA M' recognizes xy such that $xy \in L$. Since M' is a finite automaton, L' is a finite automaton and so, L' is proven to be regular.

4. Define $L(a, b) = \{1^{an+b} \mid n \in \mathbb{N}\}$. Let $L \subseteq 1^*$. Prove that L is regular if and only if there exists numbers $a_1, b_1, \dots, a_k, b_k \in \mathbb{N}$ and a finite language $F \subseteq 1^*$ such that

$$L = L(a_1, b_1) \cup \dots \cup L(a_k, b_k) \cup F$$

You are proving the unary regular languages are exactly those which are unions of arithmetic progressions, and some finite extra strings.

Answer: We wish to show for any unary language L , the following two expressions are equivalent: (1) L is regular, and (2) L can be expressed as a finite union of arithmetic progressions and some finite extra strings (F).

We begin by showing (2) $\Rightarrow$ (1). Applying the union definition to L , we have $L = \bigcup_{i=1}^k \{1^{a_i n + b_i} \mid n \in \mathbb{N}\} \cup F$, where a, b are integers and F is a finite set. We define the operations of the DFA to be the following:

- (a) The automation has states $\{0, 1, \dots, a-1\}$.
- (b) The starting state is 0.
- (c) When reading a symbol 1, the state updates such that $q \rightarrow (q+1) \bmod(a)$.
- (d) The accepting states are equivalent to $b \bmod(a)$. In Layman's terms, we wish to only accept states whose lengths are $an + b$.

Therefore, we find arithmetic progression languages are regular. Furthermore, we recognize any finite set of unary strings can be recognized by a DFA that hard-codes each string in F . Therefore, we find the finite set F to be regular. Similarly, since regular languages are closed under union, we find L to be regular. Therefore, any language that is a union of arithmetic progressions plus finite extra strings must be regular.

We then show (1) $\Rightarrow$ (2). Relatively trivially, we define $M = (Q, \{1\}, \delta, q_0, F)$. Since $|Q|$ is non-infinite, it must repeat a state (pigeonhole principal (shoutout CS 3510)). We observe there exists integers $m \geq 0$ and $r > 0$ such that the state after reading symbols of 1 is the same as the state after reading $m + r$ symbols. From this point, the automaton behaves periodically with period r .

We decompose L into two parts. (1) We begin by considering the finite initial portion (before the period of r occurs). In other words, we consider strings 1^n such that $n < m$. We individually check if these strings are in L . We define this finite set to be F_0 . (2) We then consider the periodic tail portion. For each residue $t \in \{0, 1, \dots, r-1\}$, if 1^{m+t} is in L , then every string of length $m+t+nr$ is also in L . We find this to be

true since the automaton is periodic after r symbols. Each subset $\{1^{m+t+nr} | n \geq 0\}$ is an arithmetic progression $L(r, m+t)$. Therefore, we observe this is no less than a finite union of arithmetic progressions plus a finite set F_0 . Therefore, we observe if L is regular, then its structure must eventually be periodic, so L is a finite union plus a finite set.

Therefore, we have shown $(2) \Rightarrow (1)$ and $(1) \Rightarrow (2)$. QED. *mic drop*